i.e., $\sqrt{3}a = 4r$

$$r = \frac{\sqrt{3}}{4}a$$

∴ Volume of the sphere with radius 'r'

$$= \frac{4}{3}\pi r^3$$

$$= \frac{4}{3}\pi\left(\frac{\sqrt{3}}{4}a\right)^3$$

$$= \frac{\sqrt{3}}{16}\pi a^3 \quad ...(1)$$

Number of spheres belong to a unit cell in bcc arrangement is equal to two and hence the total volume of all spheres

$$= 2 \times \left(\frac{\sqrt{3}\ \pi\ a^3}{16}\right) = \frac{\sqrt{3}\ \pi\ a^3}{8}$$

Dividing (2) by (3)

$$\text{Packing fraction} = \frac{\left(\frac{\sqrt{3}\ \pi\ a^3}{8}\right)}{\left(a^3\right)} \times 100$$

$$= \frac{\sqrt{3}\ \pi}{8} \times 100$$

$$= \sqrt{3}\ \pi \times 12.5$$

$$= 1.732 \times 3.14 \times 12.5$$

$$= 68\ \%$$

i.e., 68 % of the available volume is occupied. The available space is used more efficiently than in simple cubic packing.

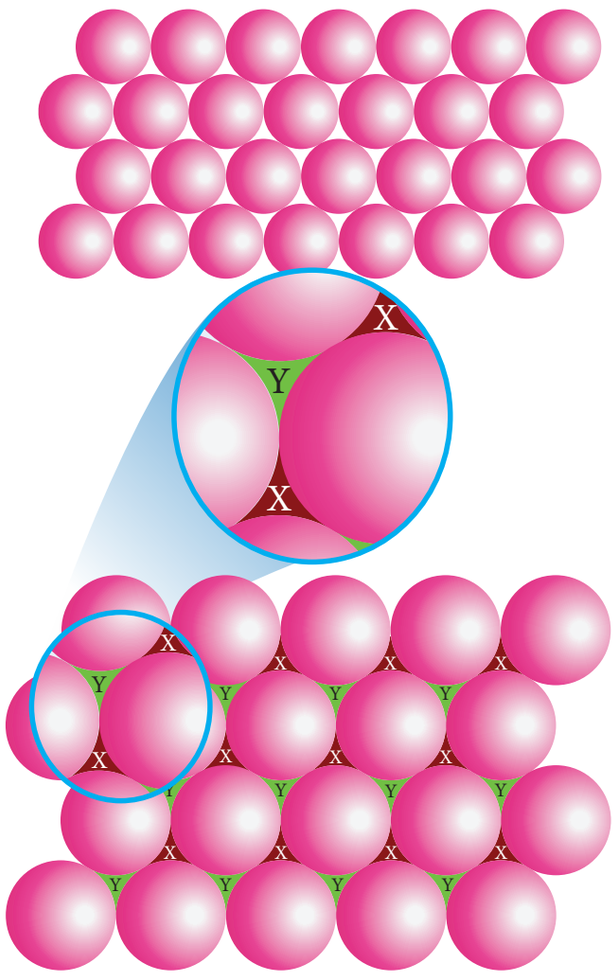
6.6.5 The hexagonal and face centered cubic arrangement:

Formation of first layer:

In this arrangement, the first layer is formed by arranging the spheres as in the case of two dimensional ABAB arrangements i.e. the spheres of second row fit into the depression of first row. Now designate this first layer as 'a'. The next layer is formed by placing the spheres in the depressions of the first layer. Let the second layer be 'b'.

Formation of second layer:

In the first layer (a) there are two types of voids (or holes) and they are designated as x and y. The second layer (b) can be formed by placing the spheres either on the depression (voids/holes) x (or) on y. let us consider the formation of second layer by placing the spheres on the depression (x).



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